ajusted means screening a fabae

instalation of all the required package

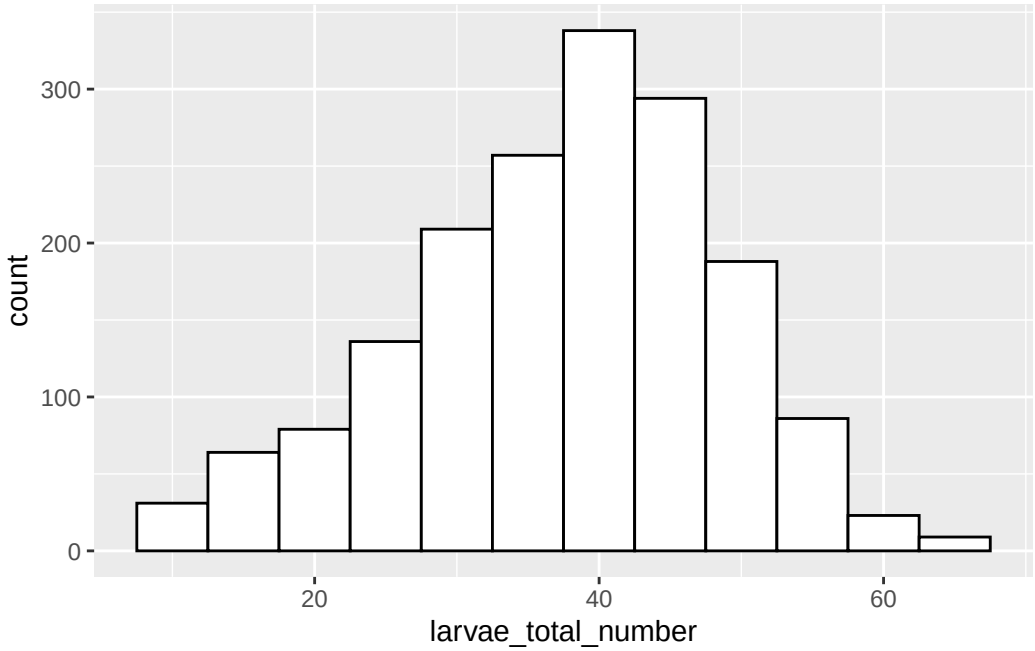
```
suppressPackageStartupMessages(library("tidyverse"))
suppressPackageStartupMessages(library("lme4"))
suppressPackageStartupMessages(library("Hmisc"))
suppressPackageStartupMessages(library("corrplot"))
suppressPackageStartupMessages(library("emmeans"))
suppressPackageStartupMessages(library("performance"))
```

preparation of the data based on the result from the filtering script :

```
data250fabae_4 <- read.csv("../../01.dataset_filtering/01_A.fabae/final filters/aphis_fabae_c

data250fabae_4 <- data250fabae_4 %>%
  mutate(ivip = ifelse(seed_batch == "KF2229", "hedin", ivip))
```

```
data250fabae_4 %>%
  ggplot(., aes(x = larvae_total_number)) +
  geom_histogram(binwidth = 5, fill = "white", color = "black")
```




```
model <- lmer(larvae_total_number ~ ivip + (1 |session/block), data = data250fabae_4)
```

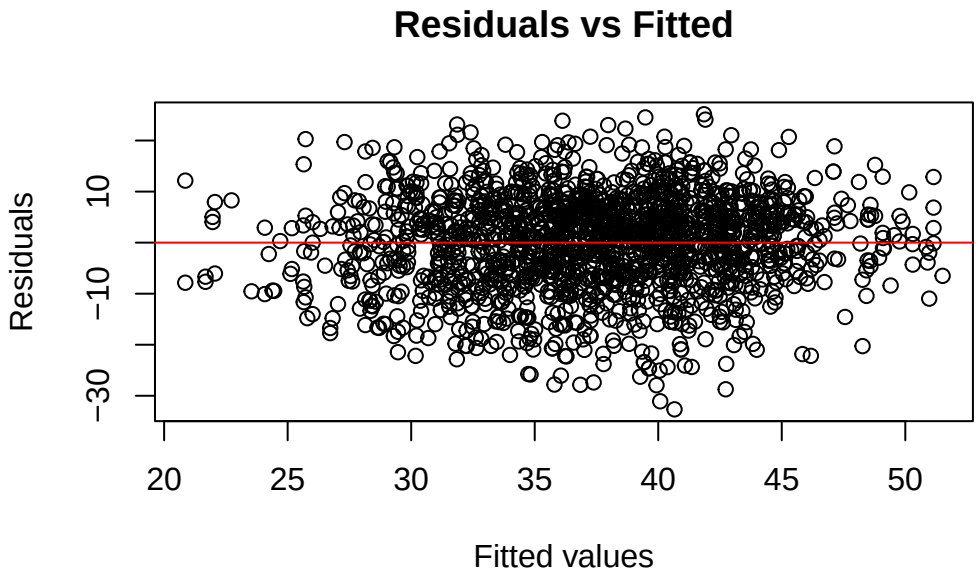
```
anova(model)
```

```
Analysis of Variance Table
     npar Sum Sq Mean Sq F value
ivip  207  31012  149.82  1.4908
```

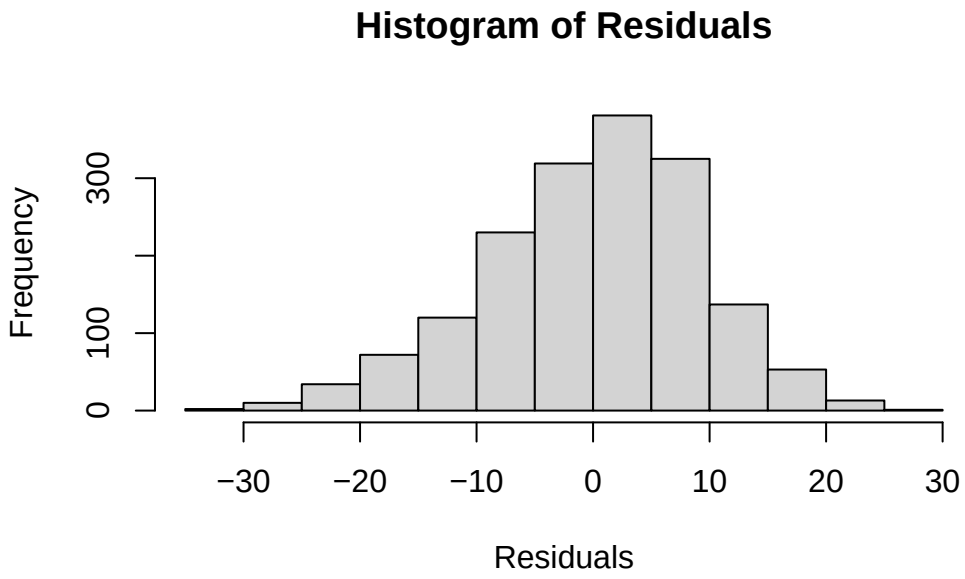
```
#variance stability

# Extract fitted values and residuals
fitted_values <- fitted(model)
residuals_values <- residuals(model)

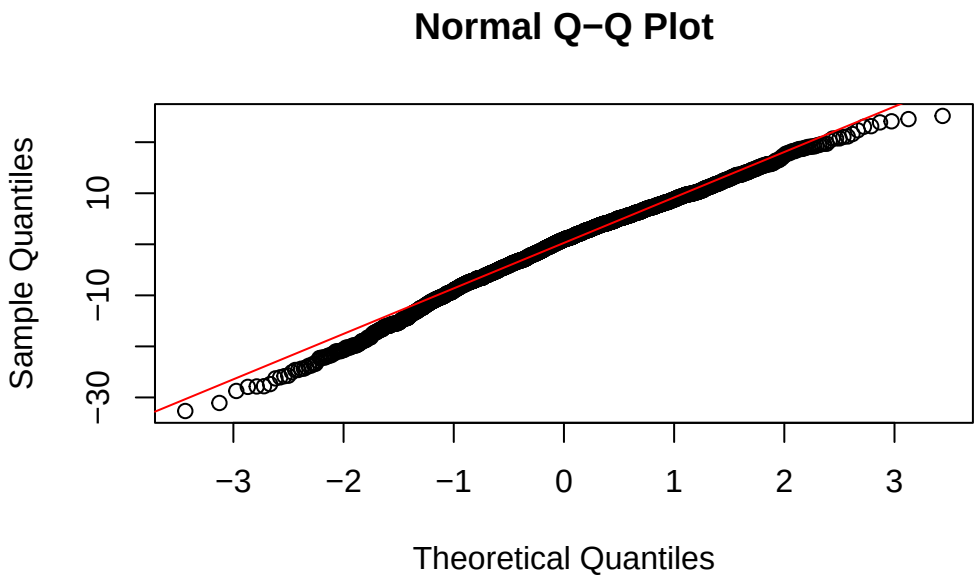
#homosedasticity
# Plot residuals vs fitted values
plot(fitted_values, residuals_values,
     main="Residuals vs Fitted",
     xlab="Fitted values",
     ylab="Residuals")
abline(h=0, col="red")
```




```r
#residual normality
hist(residuals_values, main="Histogram of Residuals", xlab="Residuals", breaks=20)
```



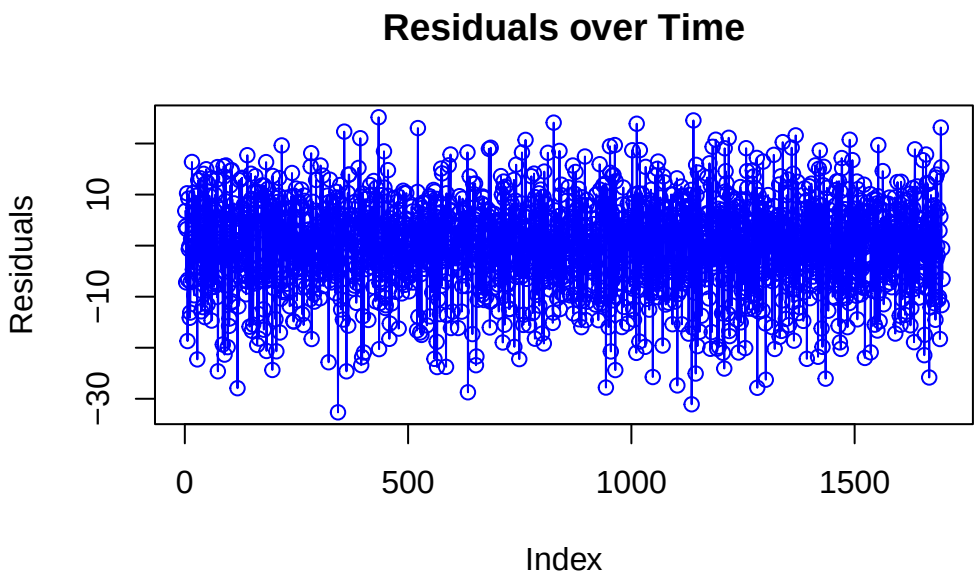
```r
# 2. Q-Q Plot for Residuals
qqnorm(residuals_values)
qqline(residuals_values, col = "red")
```




```r
shapiro_result <- shapiro.test(residuals_values)

# Print the p-value (this is your quantification of normality)
shapiro <- shapiro_result$p.value

#residual independance
# Plot residuals over time (for time-series data)
plot(residuals(model), type="o", col="blue", main="Residuals over Time", xlab="Index", ylab="
```




```
emm_genotype <- emmeans(model, ~ ivip)
emm_df <- as.data.frame(emm_genotype)



# Define the file path
file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

# Save the data frame to the specified location
write.csv(emm_df, file = file_path, row.names = FALSE)

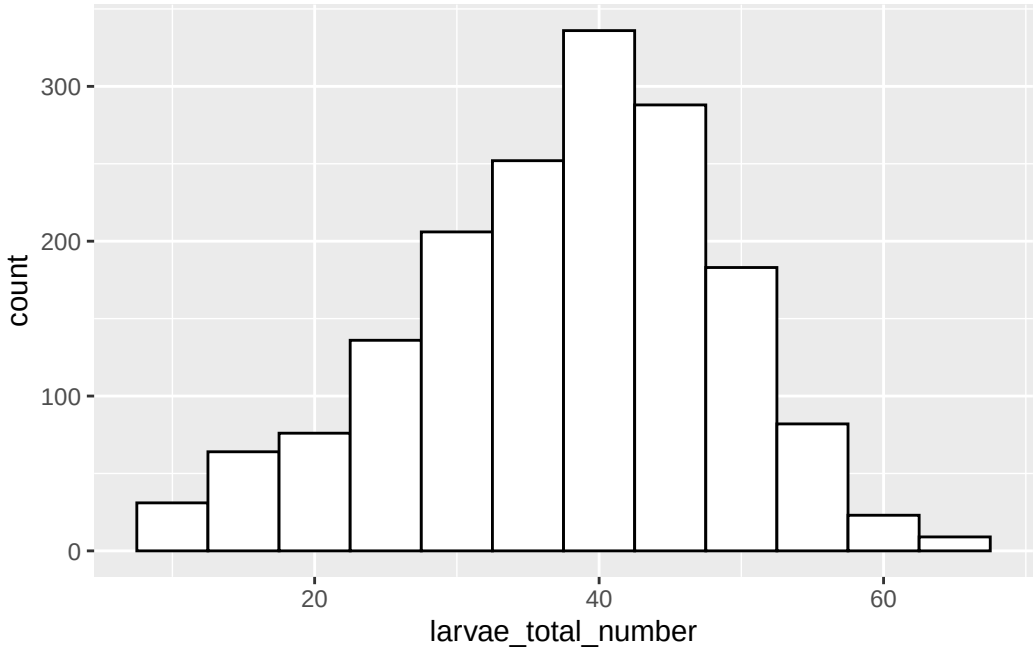
genotype_name<- emm_df[,1]

file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

write.csv(emm_df, file = file_path, row.names = FALSE)
```

```
data250fabae_5 <- read.csv("../../01.dataset_filtering/01_A.fabae/final filters/aphis_fabae_
```

```
data250fabae_5 %>%
  ggplot(., aes(x = larvae_total_number)) +
  geom_histogram(binwidth = 5, fill = "white", color = "black")
```




```
model <- lmer(larvae_total_number ~ ivip + (1 |session/block), data = data250fabae_5)
```

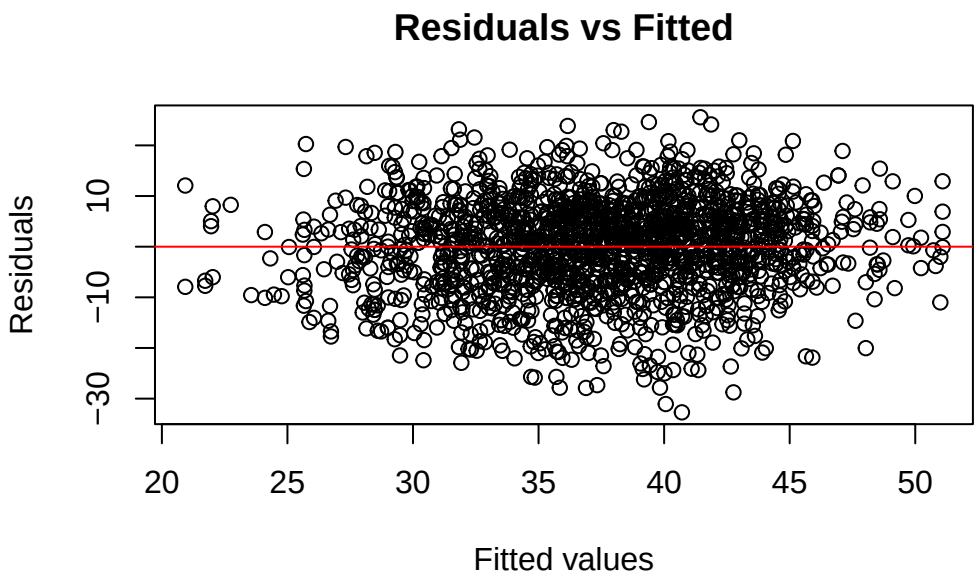
```
anova(model)
```

```
Analysis of Variance Table
     npar Sum Sq Mean Sq F value
ivip  199  29762  149.56  1.4784
```

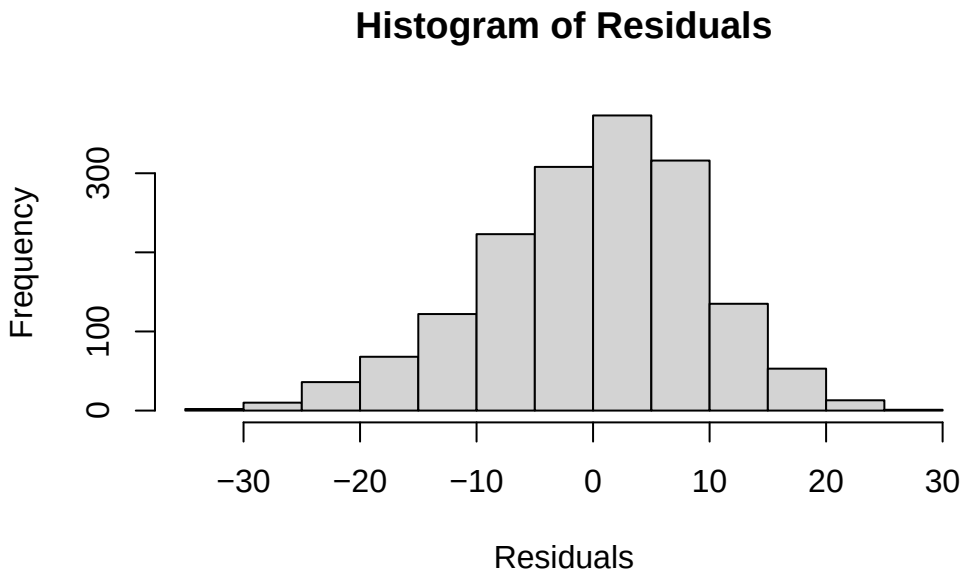
```
#variance stability

# Extract fitted values and residuals
fitted_values <- fitted(model)
residuals_values <- residuals(model)

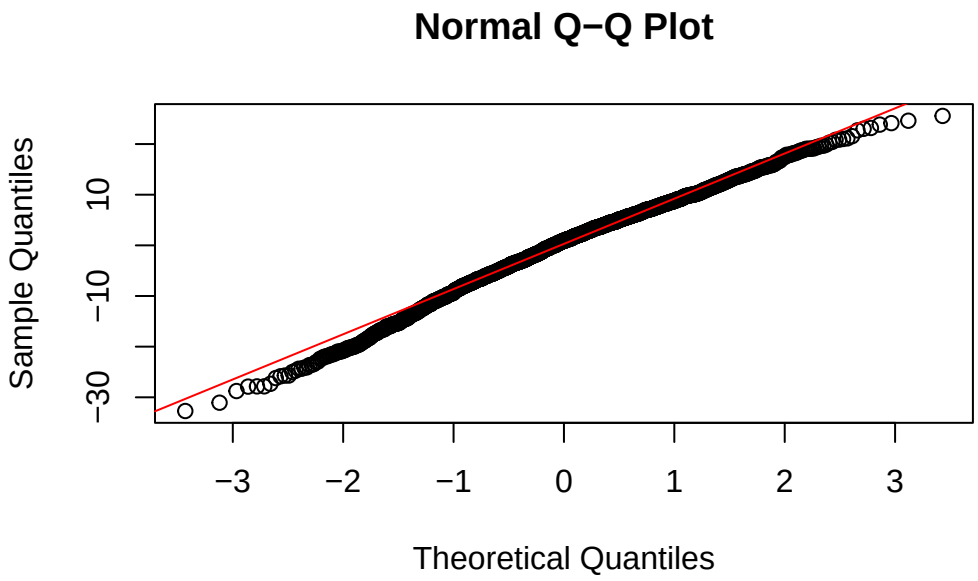
#homosedasticity
# Plot residuals vs fitted values
plot(fitted_values, residuals_values,
     main="Residuals vs Fitted",
     xlab="Fitted values",
     ylab="Residuals")
abline(h=0, col="red")
```




```r
#residual normality
hist(residuals_values, main="Histogram of Residuals", xlab="Residuals", breaks=20)
```



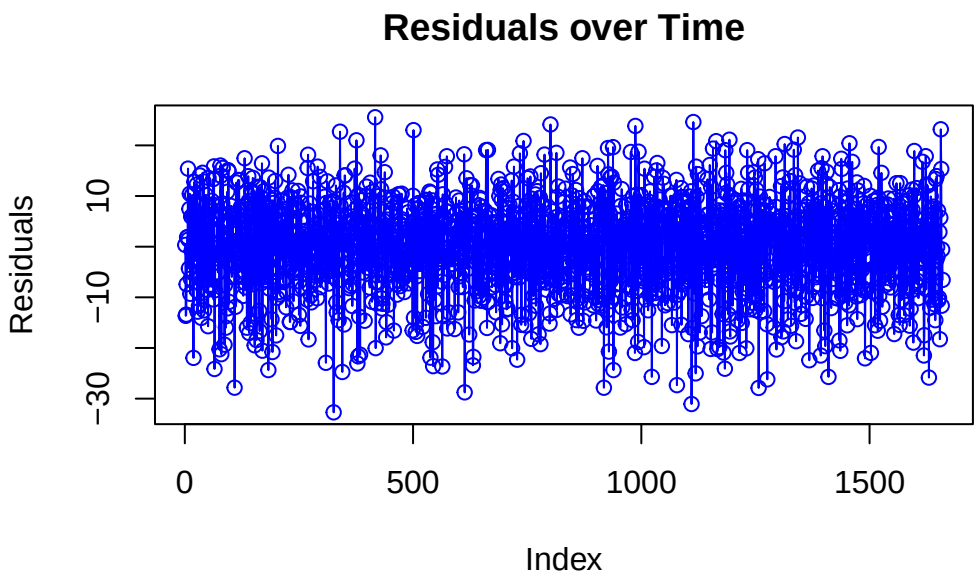
```r
# 2. Q-Q Plot for Residuals
qqnorm(residuals_values)
qqline(residuals_values, col = "red")
```




```r
shapiro_result <- shapiro.test(residuals_values)

# Print the p-value (this is your quantification of normality)
shapiro <- shapiro_result$p.value

#residual independance
# Plot residuals over time (for time-series data)
plot(residuals(model), type="o", col="blue", main="Residuals over Time", xlab="Index", ylab="
```




```
emm_genotype <- emmeans(model, ~ ivip)
emm_df <- as.data.frame(emm_genotype)



# Define the file path
file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

# Save the data frame to the specified location
write.csv(emm_df, file = file_path, row.names = FALSE)

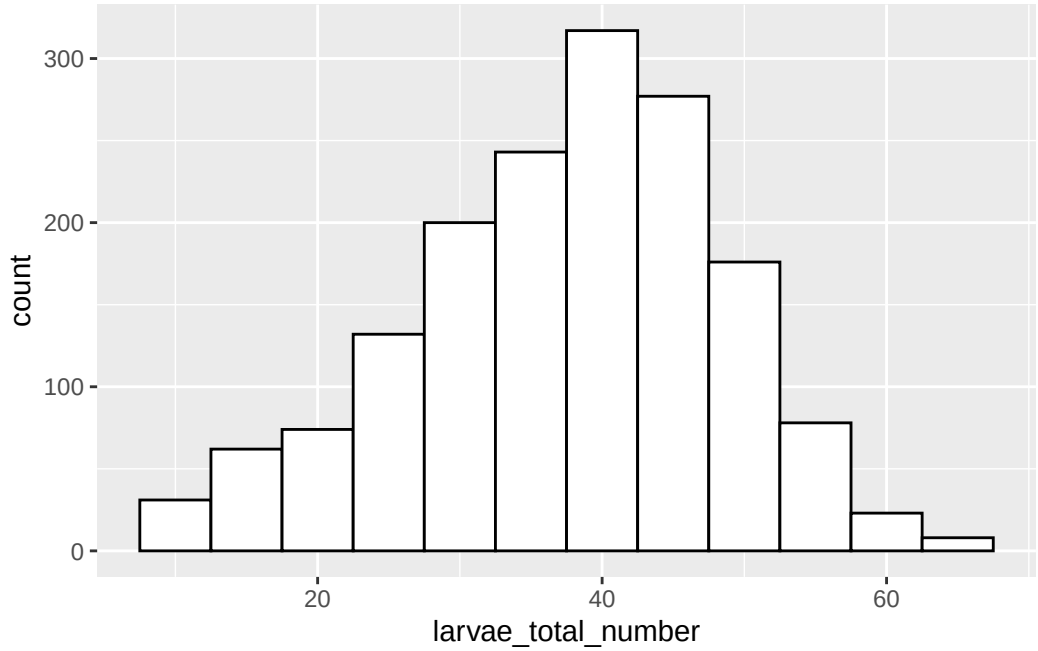
genotype_name<- emm_df[,1]

file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

write.csv(emm_df, file = file_path, row.names = FALSE)
```

```
data250fabae_6 <- read.csv("../../01.dataset_filtering/01_A.fabae/final filters/aphis_fabae_
```

```
data250fabae_6 %>%
  ggplot(., aes(x = larvae_total_number)) +
  geom_histogram(binwidth = 5, fill = "white", color = "black")
```




```
model <- lmer(larvae_total_number ~ ivip + (1 |session/block), data = data250fabae_6)
```

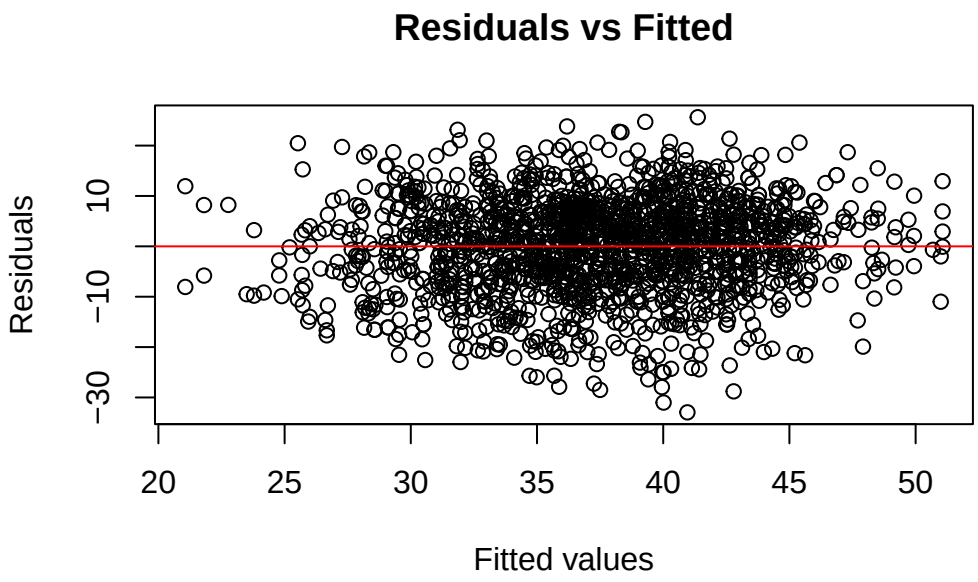
```
anova(model)
```

```
Analysis of Variance Table
     npar Sum Sq Mean Sq F value
ivip  186  27590  148.33  1.4454
```

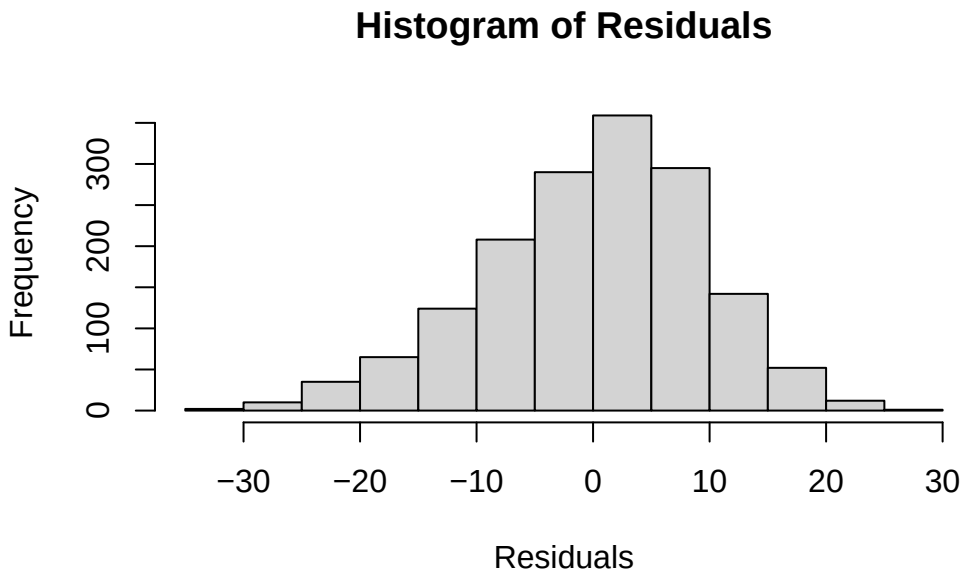
```
#variance stability

# Extract fitted values and residuals
fitted_values <- fitted(model)
residuals_values <- residuals(model)

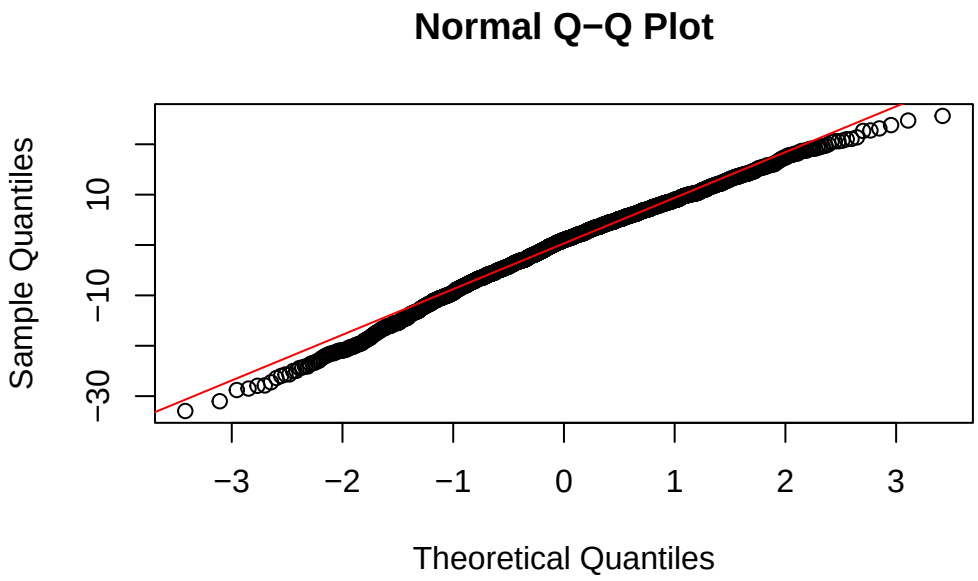
#homosedasticity
# Plot residuals vs fitted values
plot(fitted_values, residuals_values,
     main="Residuals vs Fitted",
     xlab="Fitted values",
     ylab="Residuals")
abline(h=0, col="red")
```




```r
#residual normality
hist(residuals_values, main="Histogram of Residuals", xlab="Residuals", breaks=20)
```



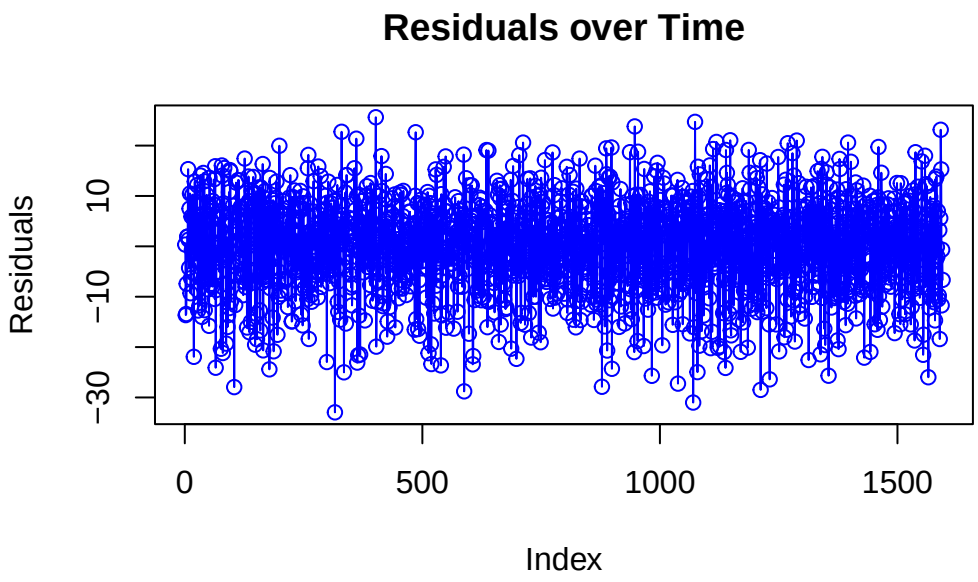
```r
# 2. Q-Q Plot for Residuals
qqnorm(residuals_values)
qqline(residuals_values, col = "red")
```




```r
shapiro_result <- shapiro.test(residuals_values)

# Print the p-value (this is your quantification of normality)
shapiro <- shapiro_result$p.value

#residual independance
# Plot residuals over time (for time-series data)
plot(residuals(model), type="o", col="blue", main="Residuals over Time", xlab="Index", ylab="
```




```
emm_genotype <- emmeans(model, ~ ivip)
emm_df <- as.data.frame(emm_genotype)



# Define the file path
file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

# Save the data frame to the specified location
write.csv(emm_df, file = file_path, row.names = FALSE)


genotype_name<- emm_df[,1]

file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

write.csv(emm_df, file = file_path, row.names = FALSE)
```